

A compressed complex network and a conditional integer-multiplication saving above 2^{-31}

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October 7, 2026

Abstract

The compact-control witness $\kappa = 83/10^{12}$ was limited by the original complex network, which spends one side wire per ordered neighbor pair. We compute the same side correction with shared sums: 108195 side roles per invocation instead of 3693800. Binary coordinate and pair-star labels give every frame transition an orthonormal residual basis, after one necessary change: no injected piece of a disjoint sum may cover every point outside its target, because that residual is an alternating hyperbolic plane. The complex saving rises from $418/10^{12}$ to $14/10^9$, and the retained paired bit network becomes the only binding motif. With the compact-control recurrence this gives the conditional witness $\kappa = 59/10^{11} > 2^{-31}$, a factor $590/83 \approx 7.11$ above the preceding witness. The underlying manuscript's interfaces remain assumptions.

1 Scope and dependencies

The source is OpenAI's *Integer multiplication below $n \log n$* , pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The computational model is a fixed finite alphabet and a fixed number of one-dimensional tapes. We retain the compact-control movement, reservation, repair and generalized-guard arguments, the paired bit network with $\tau = 1 - 296/10^{11}$, the nonadjacent routing, the Gaussian width $\alpha = \lceil (32db)^{1/4} \rceil$, and all resampling, transform, error and rounding interfaces. Only the complex motif's side circuit changes; its ground size $h = 25$, central wires and data-stage frames are those of the compact-control construction, recalled next.

2 The retained complex motif

2.1 An independently sized complex network

Keep the paired bit network at $h_b = 50$, with address-swap exponent $\tau = 1 - 296/10^{11}$. Instantiate the original complex construction instead at $h_c = 25$. Its constants, distinguished from those of the bit network, are

$$m_c = 15625, \quad v_c = 2300, \quad N_c = 12167000000,$$

$$W_c = 58645352620000, \quad L_c = 10315500000, \quad s_c = 916333630984500000.$$

Here $W_c = 2v_c^3 + 3v_c^2(v_c z_c + 26)$, $z_c = \binom{22}{3} + 66$, and $L_c = 3v_c^2 \cdot 25 \cdot 26$.

*Prepared with assistance from Claude (Anthropic). This is not independent review or endorsement by Anthropic.

Proposition 1. *The original complex phase interface, including its signed scalar bank exchange, arbitrary-scratch restoration, binary residual bases and endpoint corrections, holds with these constants. Its normalized residual saving supports the strict exponent $\sigma = 1 - 418/10^{12}$.*

Proof. The original construction and residual table apply at this ground size. For an intersection of size j , the center coefficient is $(j - 1)/2$; the side correction is its negative for distinct even-intersection neighbors. Their sum is zero for $j = 0, 1, 2$ and one for $j = 3$. The same eight-step schedule therefore restores arbitrary scratch and gives the elementary bank shear, and the three stages give the signed exchange.

Binary triple indicators have norm one. The orthogonal decompositions in the residual table remain nondegenerate. Every nonzero residual has a norm-one vector: choose a coordinate outside the support of a triple or two neighboring triples (at most six coordinates), or outside an earlier/future tensor indicator (support $3^j < 25^j$); tensor with the remaining norm-one factors. Thus the original nonalternating-space argument supplies the required orthonormal bases. No even-ground-size matching or rational bit-interface positivity is needed for this unchanged complex network.

The rank accounting gives

$$s_c = W_c m_c - 2N_c + 2L_c, \quad \eta_c = \frac{W_c m_c - s_c}{W_c m_c} = \frac{14}{3464399375} > 0.$$

The needed positivity is $L_c < N_c$, not $2L_c < N_c$. Endpoint weight-27 phase corrections and the signed-exchange correction are unchanged. Exact rational logarithm bounds give $\log m_c < 966/100$ and $\eta_c > (418/10^{12})(966/100)$. The inequality $e^{-x} > 1 - x$ then yields $s_c/W_c < m_c^\sigma$. The certificate also checks $2 \leq s_c < m_c^5$. $\square$

The bit and complex dimensions need not agree. The bit interface supplies swaps at arbitrary binary widths. The complex recursion uses only that interface's exponent τ and fixed tape guarantee, while its own branching, role split and depth use m_c, W_c, s_c . All occurrences of m, W, s within the revised layer proof denote these complex constants. The retained assembly consumes their time exponents and completed-layer contract, so it introduces no equality requirement between the two arities.

3 The compressed side circuit

3.1 A compressed side circuit for the complex network

Keep the complex motif of Section 2.1: $h = 25$, $v = \binom{25}{3} = 2300$, $N = v^3$, $m = h^3 = 15625$, scalars in $\mathbb{Z}[i, 1/2]$, labels in $F = \mathbb{F}_2^{25}$ with the dot product, the $h + 1$ central wires, and the gather and scatter maps. For a triple S write t_S for its indicator. The original side wires add

$$\frac{1}{2} \sum_{T \cap S = \emptyset} x_T - \frac{1}{2} \sum_{|T \cap S|=2} x_T \tag{1}$$

to y_S . With the central coefficient $(|S \cap T| - 1)/2$ this gives $y \leftarrow y + x$. They use one wire per ordered neighbor pair, that is $v z_c = 3693800$ side wires per invocation. We compute the same map with shared sums.

All sums below are formal sums of inputs x_T whose two summands have disjoint supports, so a source contributes to a node exactly when a directed path joins them. Equal supports are identified within each of the two families below, and nodes that feed no injected piece are removed.

Disjoint sums. Order the points $0, \dots, 24$. For a list of nodes indexed by distinct points and $k \leq 3$, the routine `vec` returns, for every set E of at most k of these points, the sum of the nodes whose points avoid E . It splits the list at $\lfloor n/2 \rfloor$, recurses, and forms each output by one addition of a left and a right output whose exclusion sets have total size at most k . For a point list X and weights $w(u, v)$ on its pairs, the routine `pairs` returns the sums over pairs avoiding E , $|E| \leq 3$: split X into halves L, R , recurse on both, apply `vec` to each cross row $(w(u, r))_{r \in R}$, then apply `vec` down the columns of these row outputs with the remaining exclusion budget, and add the three disjoint regions. The routine `tri` does the same for triple sums: with halves L, R of X , put

$$P_r = \text{pairs}(L, (u, v) \mapsto x_{uvr}), \quad P'_u = \text{pairs}(R, (a, b) \mapsto x_{uab}),$$

apply `vec` over $r \in R$ to $P_r[E_L]$ and over $u \in L$ to $P'_u[E_R]$, and add the four disjoint regions: triples in L , triples in R , two points in L with one in R , and one point in L with two in R .

At the top level $L = \{0, \dots, 11\}$ and $R = \{12, \dots, 24\}$. For a target S with $S_L = S \cap L$ and $S_R = S \cap R$ the disjoint pieces are

$$\text{tri}(L)[S_L], \quad \text{tri}(R)[S_R],$$

the two `vec` outputs over the halves $R_1 = \{12, \dots, 17\}$ and $R_2 = \{18, \dots, 24\}$ of the family $P_r[S_L]$, and the two `vec` outputs over the halves $L_1 = \{0, \dots, 5\}$ and $L_2 = \{6, \dots, 11\}$ of the family $P'_u[S_R]$. These six pieces partition the triples disjoint from S . They are internal nodes of the full computations because `vec` splits at the same halves.

Pair stars. For a pair $a < b$, list the other points as $u_1 < \dots < u_{23}$ and form prefix sums $\pi_i = \sum_{j \leq i} x_{abu_j}$ and suffix sums $\sigma_i = \sum_{j \geq i} x_{abu_j}$. Target $\{a, b, u_i\}$ receives π_{i-1} and σ_{i+1} . Over its three pairs, S receives exactly the triples meeting it in two points.

Injection rule. Inject every disjoint piece with coefficient $1/2$ and every pair-star piece with coefficient $-1/2$, subject to one rule: if the triples of a piece cover every point outside S , inject its two summands instead, recursively. A single triple covers three of the 22 outside points, so the rule terminates. At $h = 25$ it never splits a disjoint piece. It splits π_{22} for $i = 23$ and σ_2 for $i = 1$. Each target therefore receives twelve pieces.

The circuit has 80595 additions (68595 disjoint, 12000 pair-star) and 27600 injected pieces. Exact support comparison checks every coefficient of (1): all 3693800 nonzero entries, their signs, and all required zeros.

3.2 The transparent schedule

Allocate reversible roles as for the bit network: a node's outgoing edges are its later uses and its injected pieces; an input slot fans out, and an addition reuses its first incoming role as a pivot and retires the second. This uses $R = c + q = 80595 + 27600 = 108195$ roles. Let L be the product of the node mixers, V the copy into input slots, and J the injection, which adds $\pm \frac{1}{2}$ times a piece role to its target. With $G, \mathcal{R}$ the original gather and scatter, a forward invocation performs

$$L, -J, L^{-1}, -\mathcal{R}, V, G, \mathcal{R}, L, J, L^{-1}, G^{-1}, V^{-1}.$$

For initial side values z and central values w , the target changes by

$$-JLz - \mathcal{R}w + \mathcal{R}(w + Gx) + JL(z + Vx) = (JLV + \mathcal{R}G)x = x,$$

and the last four operations restore z and w . The inverse schedule, with logical source Y and target X , gives $X \leftarrow X - Y$; forward, inverse and forward invocations send (X, Y) to $(-Y, X)$ and restore every auxiliary value. Every gate is pointwise linear with coefficients in $\{0, \pm 1, \pm \frac{1}{2}\}$.

3.3 Binary labels

Assign each node z a subspace $U_z \subset F$:

- an input node T receives $\langle t_T \rangle$;
- a disjoint-sum node receives the coordinate space $F_X = \langle e_x : x \in X \rangle$, where X is the union of its triples;
- a pair-star node for $\{a, b\}$ with leaf set Λ receives $\langle t_{abu} : u \in \Lambda \rangle$.

The vectors t_{abu} have weight three and pairwise intersections of size two, so they form an orthonormal basis of the pair-star label. Coordinate spaces have orthonormal coordinate bases. Every label is therefore nondegenerate. Along a dataflow edge $z \rightarrow w$ the support grows, so $U_z \subset U_w$. The injection gate for target S uses the physical Y frame, whose active component is $t_S^\perp$. Every triple of a piece meets S in an even number of points, so the piece label lies in $t_S^\perp$.

Lemma 1. *If a piece leaves some point $y \notin S$ uncovered, then the residual $t_S^\perp \cap U_z^\perp$ of its injection edge has an orthonormal basis. If instead a disjoint piece covers every point outside $S = \{a, b, c\}$, its injection residual is $\langle e_a + e_b, e_b + e_c \rangle$, which has none.*

Proof. Both U_z and $t_S^\perp$ are nondegenerate, so the residual is nondegenerate. The vector e_y is orthogonal to t_S , to every coordinate vector in a disjoint label, to every t_{abu} with $u \neq y$, and to an input line t_T with $y \notin T$. Thus the residual contains a vector of norm one and is nonalternating; the manuscript's absorption argument then gives an orthonormal basis. In the second case $t_S^\perp = F_{[h] \setminus S} \perp \langle e_a + e_b, e_b + e_c \rangle$. The displayed plane is nondegenerate, and all its vectors have even weight, hence norm zero. $\square$

A nondegenerate label between $F_{[h] \setminus S}$ and $t_S^\perp$ must be one of these two spaces, since a proper nonzero part of the plane is an isotropic line. A full-coverage disjoint piece thus cannot be repaired by relabeling; the injection rule avoids it. Any partition of the disjoint triples into pieces that each miss an outside point needs at least four missed points, so four pieces per target are necessary for this label class.

The remaining residuals are covered by the following list, which the checker verifies by exact binary linear algebra for every node, edge, piece and compiled role.

- Coordinate to coordinate: $F_{X' \setminus X}$, a coordinate space.
- Input line to coordinate space: $F_X \cap t_T^\perp$ contains e_x for $x \in X \setminus T$, which is nonempty for an addition node.
- Pair-star chain: the span of the new t_{abu} .
- Entry from the early frame: the label itself.
- Exit to the late frame: $U_z^\perp$, which contains a unit coordinate vector outside the covered points; for a pair-star node, outside $\{a, b\} \cup \Lambda$, which omits at least the excluded third point.

Each nonzero residual is nondegenerate and contains a vector of odd weight, so it has an orthonormal basis.

3.4 Frames in both stage directions

Use the frames of the bit construction with these binary labels. With $A = B \perp P$ and future line factor Q , put $D_0 = B \otimes F$, $D_1 = A \otimes F$ and $D_U = (B \otimes F) \perp (P \otimes U)$, all tensored by Q . Early mixers and the early injection use D_0 ; middle node gates use D_{U_z} ; copy gates use the physical X frame $D_{\langle t_T \rangle}$ and injection gates the physical Y frame $D_{t_S^\perp}$; late cleanup uses D_1 . An edge residual is $P \otimes E \otimes Q$ for a residual $E \subset F$ from the list above. The lines P, Q are spanned by tensor products of triple indicators, which have norm one, so tensoring an orthonormal basis of E gives one of the residual. The data wires pass through the same frames as in the original motif and have the same residuals.

In the inverse schedule the nontrivial labels occur in the first L^{-1} . Assign $D_{U_z^\perp}$ to node z there, D_0 to the preceding L , and D_1 to the second L , J and L^{-1} . A forward inclusion $U_z \subset U_w$ gives $U_w^\perp \subset U_z^\perp$ with the same residual $U_w \cap U_z^\perp$. A piece role enters from the physical X line $\langle t_S \rangle \subset U_z^\perp$ with residual $t_S^\perp \cap U_z^\perp$, the forward injection residual. An input role starts at the physical Y frame $B \otimes \langle t_T \rangle$ of the first copy, passes D_0 with residual $B \otimes t_T^\perp$ as a data wire does, and ends at $t_T^\perp$, exactly its physical Y frame. Entries and exits exchange the forward residuals U_z and $U_z^\perp$. Thus every residual of the inverse stage is one already shown to have an orthonormal basis.

3.5 Rank, phases and depth

Proposition 2 (Compressed complex interface). *The complex network with this side circuit has*

$$\begin{aligned} W_c &= 2N + 3v^2(R + h + 1) = 1741801270000, \\ L_c &= 3v^2h(h + 1) = 10315500000, \\ s_c &= W_c m - 2N + 2L_c = 27215641140750000, \end{aligned}$$

satisfies the binary phase interface of Proposition 1 with these constants, and has $\eta_c = (W_c m - s_c)/(W_c m) = 28/205789375$.

Proof. The schedule gives the signed exchange and restores arbitrary scratch. Every side role has source label 0 and sink label $F^{\otimes 3}$, and its labels increase, so it contributes exactly m to the residual count. Data wires and central wires are unchanged, so the only decreases remain the h -dimensional central returns. Telescoping gives s_c . Every residual has an orthonormal basis, so each edge is a product of one translation kernel per basis vector, as in the original phase interface. The endpoint phases of the data roles and the signed-exchange correction are unchanged, and every auxiliary role still runs from frame I to $C^{\otimes m}$.

The coefficient-depth argument needs a bound on gates, not on gates per wire. An invocation has $4 \cdot 82895$ mixer gates, $2v$ copy gates, $2v$ injection gates (one per target) and four central gates. Over $3v^2$ invocations this is $5408242080000 \leq 12W_c$. Each gate has at most W_c inputs and outputs and coefficients in $\{0, \pm 1, \pm \frac{1}{2}\}$, so $E = 64(W_c + m + 1)^3$ bounds the depth of one invocation as before. Also $2 \leq s_c < m^5$, so the generalized stopped guard applies with these constants. All gates, labels and bases are fixed finite data. $\square$

The rational log bound $\log m < 966/100$ and $\eta_c > (14/10^9)(966/100)$ give, by $e^{-x} > 1 - x$,

$$s_c/W_c < m^\sigma, \quad \sigma = 1 - \frac{14}{10^9}. \quad (2)$$

This is 33.49 times the previous complex saving $418/10^{12}$.

4 Parameters and assembly

Put $a = 296/10^{11}$ and $a_c = 14/10^9$. Choose

$$\begin{aligned} \tau = 1 - a, \quad \sigma = 1 - a_c, \quad \epsilon = \frac{19999}{100000}, \quad c = \frac{999}{1000}, \quad \beta = \frac{1}{1000}, \quad \zeta = \frac{1}{10000}, \quad \delta = \frac{1}{10^6}, \\ C_1 = 5 - 4\beta + \zeta = \frac{49961}{10000}, \quad \lambda = 1 - \frac{2958}{10^{12}}, \quad \lambda' = 1 - \frac{2956}{10^{12}}, \quad \kappa = \frac{59}{10^{11}}. \end{aligned}$$

Now $\sigma < \tau$, so the compact-control internal exponent is $\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\} = \tau$. Exact comparisons give

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c\} < \lambda', \quad \epsilon C_1 = \frac{999170039}{10^9} < 1.$$

The compact-control layer theorem therefore supplies the completed normalized-butterfly contract at cost $O(Vd^{\lambda'})$, using the complex constants of Proposition 2 and the generalized guard with $B = s_c + 64(W_c + m + 1)^3$.

The spacing exponent enters only through $\epsilon(1 + c) < 1$, which gives $K = \Theta(p^{\epsilon c}) = o(\ell)$, through $K/\log p \rightarrow \infty$, and through the reserved-axis count $O(\log d + d^{1-c} \log p) = o(d)$. All three hold. The retained Gaussian setup needs $\epsilon < 1/5$ for the cutoff $46b^{(1+3\epsilon)/2} \leq b/4$ at $b \geq 2^{40}$, and gives $\alpha = \Theta(p^{119999/400000})$ and $\gamma = O(p^{159997/200000}) = o(p)$. Prime selection has $1 - 2\epsilon = 30001/50000 > 0$.

The seven assembly margins keep their form:

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), \\ g_3 &= \epsilon(1 - \lambda'), & g_4 &= (1 - \tau)(1 - \epsilon), \\ g_5 &= 1/4 - \delta - 5\epsilon/4, & g_6 &= 1 - \delta - \epsilon, \quad g_7 = \epsilon. \end{aligned}$$

The former choice $c = 1/5$ would now make g_2 binding; with $c = 999/1000$, exact arithmetic gives

$$G_* = \min_i g_i = g_3 = \frac{14779261}{25 \cdot 10^{15}}, \quad G_* - \kappa = \frac{29261}{25 \cdot 10^{15}} > 0.$$

This strict gap absorbs all remaining fixed powers of $\log p$, and the retained assembly yields

$$T(n) = O(n(\log n)^{1-59/10^{11}}), \quad 2^{-31} < \frac{59}{10^{11}} < 2^{-30},$$

conditional on the identified upstream and extension proofs.

The next ceiling. With $\sigma < \tau$ the layer needs $\lambda > \tau$, so $g_3 < \epsilon(1 - \tau) = \epsilon a$, and the Gaussian margin forces $\epsilon < 1/5$. For the published bit network and these assembly inequalities,

$$\kappa < \frac{a}{5} = \frac{296}{5 \cdot 10^{11}} < 2^{-30}.$$

The witness is above 99.6% of this bound. The complex motif now has spare saving: a_c exceeds a by a factor 4.7. Further progress in this strategy therefore needs a stronger bit network, or a Gaussian step permitting larger ϵ . This is a scoped bound, not a ceiling for other networks or algorithms.

5 Reproducibility and review boundary

The artifacts are

- `scripts/complex_circuit.py`: the circuit, injection rule, binary label algebra, reversible role compiler, both-direction role checks and an exact scalar simulator;
- `scripts/complex_network.py`, with certificate
`certificates/complex-network.json`: the full $h = 25$ checks, counts, guard, gate bound and parameter witness;
- `tests/test_complex_network.py`: label algebra against enumeration, the alternating-plane obstruction, full small cases ($h = 8, \dots, 12$) with dirty scratch, inverse invocations, the signed exchange and every input basis vector, and negative parameter cases.

The full check compares all 3693800 nonzero side coefficients and all zeros, checks all 82895 labels and 188790 inclusions with their residuals, and follows all 108195 compiled roles in both directions. These checks support the written arguments; they are not a formal verification of the upstream theorem, and the new construction has not received independent review.